

RTS Game Design Document

Overview

An offline hero/unit designer tool for an RTS (Real-Time Strategy) game. The tool manages hero data through a web interface backed by Excel workbooks, with support for custom fields, categories, images, and versioned design documents.

Core Hierarchy

Kingdom → Age → Hero → Stats/Properties

- Kingdom**: Race-based faction (e.g., Humans, Dwarves, Elves)
- Age**: Technological/civilizational era (e.g., Stone Age, Medieval Age)
- Hero**: Individual unit with stats, abilities, and images
- Stats**: Dynamic properties grouped by category

The Seven Kingdoms

#	Kingdom	Dominant Races	Description
1	Humans	Human	The baseline kingdom, versatile and balanced
2	Kingdom of Ores	Dwarves	Mountain-dwelling master craftsmen and warriors
3	Kingdom of Magic	Elves, Wizards, Gnomes	Arcane-focused, ranged and spellcasting units
4	Kingdom of Evil	Necromancers, Witches, Goblins	Dark magic, undead, swarms, debuffs
5	Kingdom of the Ancients	Silurian, Dinosaurs	Primitive but powerful, riding dinosaurs
6	Kingdom of the Beasts	Minotaurs, Trolls, Balrogs, Dragons	Sensuous creatures, brute strength, terror
7	Kingdom of the Divines	Gods, Angels, Demigods, Oracles, Archangels	Divine beings, holy powers, extremely powerful

Each kingdom progresses through all Ages, but their units reflect the kingdom's thematic identity within each era.

Ages (Technological Progression)

All kingdoms share the same timeline:

- Classical Age** — Basic civilization, early warfare
- Medieval Age** — Knights, castles, organized religion
- Mythic Age** — Divine intervention, legendary heroes, magic peaks

Note: Future ages may include Stone Age, Bronze Age, Iron Age, etc.

Hero System

Properties

Heroes have the following default properties, all editable and extensible:

Category	Fields
Identity	Hero Name, Kingdom, Age, Temple, God, Role, Unit Type
Combat Stats	Health, Mobility, Melee ATK, Ranged ATK, Magic ATK, Dodge, Parry
Defense	Melee DEF, Ranged DEF, Magic DEF
Abilities	Passive, Abilities, Ultimate Skill
Resource	Cost, Population
Notes	Free text notes, strengths/weaknesses

All property fields are dynamically configurable — add, rename, delete, or recategorize any field.

Roles

Example roles by kingdom:

Kingdom	Common Roles
Humans	Knight, Archer, Pikeman, General
Kingdom of Ores	Ironbreaker, Runesmith, Slayer, Engineer
Kingdom of Magic	Arcanist, Ranger, Enchanter, Druid
Kingdom of Evil	Necromancer, Warlock, Goblin Shaman, Witch
Kingdom of the Ancients	Dino-Rider, Beastmaster, Shaman, Chieftain
Kingdom of the Beasts	Minotaur Berserker, Dragon, Troll Brute, Balrog
Kingdom of the Divines	Paladin, Oracle, Archangel, Demigod

Sample Heroes

Paladin (Kingdom of the Divines, Mythic Age)

- Role: Heavy Tank
- Health: Extremely High
- Melee ATK: Very High
- Passive: Divine Retribution (counterattack every hit)
- Ultimate: Bulwark of Athenia
- Strong vs: Archers | Weak vs: Mages

Sword Master (Kingdom of Magic, Mythic Age)

- Role: Duelist
- Health: Moderate
- Mobility: High
- Melee ATK: Extremely High
- Passive: Battle Focus
- Ultimate: Barathor's Wrath

Game Mechanics

Rage System

- Heroes build Rage through combat
- Required to cast Ultimate Skills
- Example: Paladin needs 1000 Rage, Sword Master needs 800

Counter System

- Heavy Tanks strong vs Archers
- Duelists strong vs other melee
- Mages strong vs Tanks (low magic defense)
- Archers strong vs flying/magic users

Tool Features

Feature	Description
Hero Editor	Visual form-based editor with portrait upload
Age Management	Create/rename/delete ages, move heroes between them
Kingdom Filter	Select a kingdom to view all its heroes across ages
Compare Mode	Side-by-side comparison of 2+ heroes with diff highlighting
Schema Manager	Add/edit/delete fields, categories, roles, gods, kingdoms
Image Support	Portrait + up to 5 extra images per hero
Excel Export	Per-kingdom sheets (e.g., `K_Humans`, `K_Kingdom_of_Ores`)
Auto-save	Changes saved every 500ms and every 10 minutes
Search	Filter heroes by name, god, temple, role, or ability text

Storage Format

The tool generates an Excel workbook with the following sheets:

Sheet	Content
`Kingdoms`	List of all kingdoms
`Ages`	List of all ages
`Fields`	Schema definition (name, category, type)
`Categories`	Field category groups
`Roles`	List of unit roles
`Gods`	List of gods/temples
`K_<Kingdom>`	Heroes per kingdom (one sheet each)

`Heroes`	Unified backup of all heroes
`Legacy_Export`	Flat table for external tools

Images are stored in data/images/ as PNG/JPG files referenced by filename.

Versioning

Design documents in the design/ folder are versioned via Git. Every save creates a commit with a descriptive message, allowing:

- Viewing change history
- Diffing between versions
- Reverting to previous designs
- Collaborative editing through GitHub

Future Extensions

- [] Unit trees / upgrade paths per kingdom
- [] Resource costs and build times
- [] Technology trees per age
- [] Terrain and weather modifiers
- [] Campaign/story integration
- [] Multiplayer balance metrics